President's Office

REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT

ARUSHA DISTRICT COUNCIL

FORM FOUR PRE-MOCK EXAMINATION – MARCH 2020

033 BIOLOGY

Time:3Hours **Tuesday,31st March 2020 a.m**

INSTRUCTIONS:

1. This paper consists of Sections A, B and C with a total of **fifteen (15)** questions.
2. Answer **all** questions from section A and B and **ONLY TWO** question from section C of which question **13** is **compulsory**.
3. Write your **examination number** at the top right hand corner of every page.
4. All writing must be in blue or black ink. Diagrams must be drawn in pencil and labeled in ink.
5. Cellphones and any unauthorized materials are not allowed in the examination room.

SECTION A(15 Marks)

Answer **all** questions in this Section

1. For each of the items (i) – (x) choose the correct answer from among the given alternatives and write its letter beside the item number.

(i) Fertilization of human eggs most often takes place in the:

A. Ovary B. Vagina

C. uterus D. Oviduct [ ]

(ii) The main aim of respiration is:

A. To release energy

B. Intake of oxygen

C. Fermentation of Glucose

D. Release of carbondioxide [ ]

(iii) A person stretched out his arm to the left to cause the motion of the arm, his:

A. Biseps contracted white triceps relaxed

B. Biseps and triceps both relaxed [ ]

C. Triceps contracted while biseps relaxed

D. Biseps and triceps did not contract or relax

(iv) Among the following, which of the organism is a protist?

A. Earth warm B. Euglena

C. Tsetse fly D. Lice [ ]

(v) A colour blind woman marries a man with normal vision. What percentage of colour blind and normal children would be expected from such a marriage?

A. 100% colour blind

B. 50% colour blind and 50% normal

C. 25% normal and 75% colour blind

D. 100% normal [ ]

(vi) Which of the following insects show incomplete metamorphasis?

A. Mosquito B. Butterfly

C. Grasshopper D. Moth [ ]

(vii) Which of the following does not show homology?

A. Beaks in birds

B. Foot structure in birds

C. Pentadactyl structures in vertebrate

D. Wings in birds and insects [ ]

(viii) During cold weather shivering occurs in a mammalian body in order to:

A. Generate heat B. Shake off coldness

C. produce sweat D. Relax the muscles [ ]

(ix) Ferns belong to division:

A. Filicinophyta B. Bryophyta [ ]

C. Angiospermophyta D. Coniferophyta

(x) The name of red pigment with iron found in the red blood cells is:

A. Myoglobin B. Bile

C. Haemoglobin D. prothrombin [ ]

2. Match the responses in LIST B with the word/phrases in LIST A by writing the letter of the correct response beside the item number:

i. Coelomate. ii. Fasciola iii. Eukaryote iv. Bacillus v. Pteridophytes	L.Trematoda M.fern plant N.body cavity O.plant P.Rod-shaped bacteria Q.Moss Plant R.Bacteria S.Body Segmented T.Cestoda U.Spherical-Shaped Bacteria

SECTION B(60 Marks**)**

Answer ***ALL*** questions in this section:

3. (a) (i) What is breathing?

(ii) List down four roles of the alveoli

(b) Explain why it is better to breath through your nose than it is to breath through your mouth.

4. (a) (i) Distinguish between phototropism and geotropism.

(ii) Name the hormone responsible for tropic response in plants.

(b) What are the effects of the following hormones in man?

(i) Insulin

(ii) Antidiuretic Hormane (Vasopressin).

5. (a) Define the following:
   (i) Nutrition (ii) Growth
   (b) Outline four functions of protein in human body.
   (c) In a practical lesson a candidate has been provided with food sample X fill the table below:

Test for	Procedures	observation	Inferences
Starch	(i) …… ……	Food sample X turned blue black	(ii) …… ……
(iii) ……. ……	To a food sample X, Potassium hydroxide was added and drops of Copper sulphate solution was added	(iv) …… ……	Protein was present

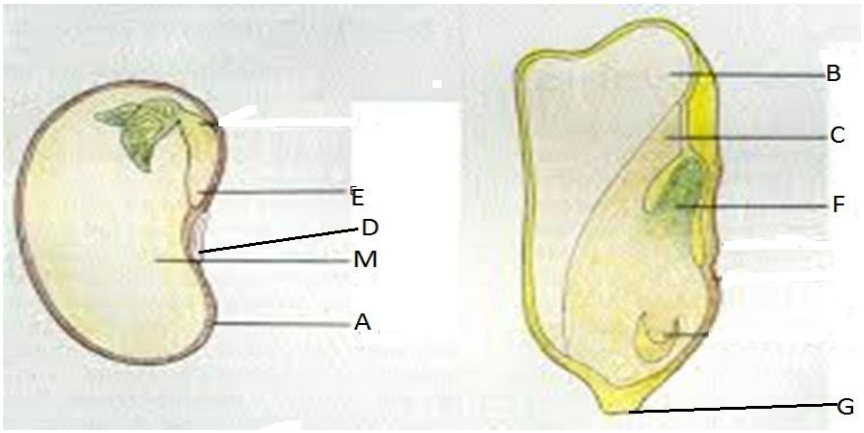
6. (a) Differentiate the following terms as used in Genetics (with examples).
   (i) Incomplete dominance and codominance
   (ii) Sex Linkage and Sex limited

   (b) List down two causes of variation among organisms.

7. (a) Name structures responsible for Gaseous exchange in:
   (i) Mammal's (ii) Young tadpole
   (iii) An insect (iv) Tilapia

   (b) Name the Plant structure concerned with Gaseous exchange (with the name of the cell responsible).

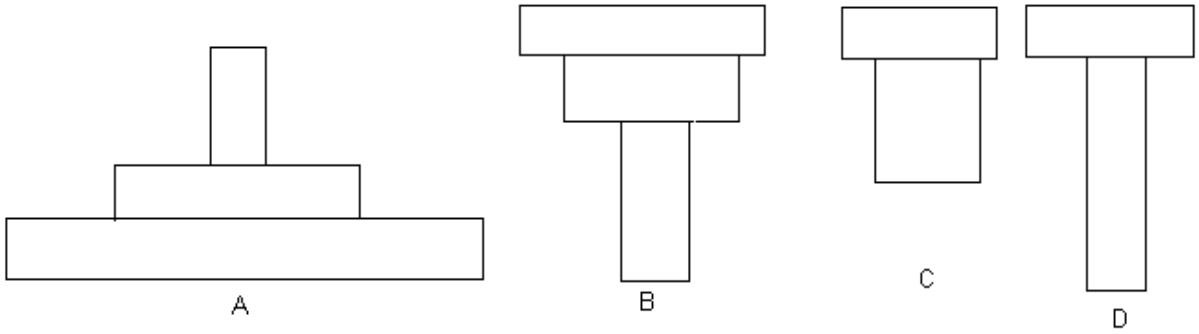
8. Observe the following diagram then answer questions that follows:



(a) Name part A – G
(b) After the development of E and F, what new plant parts will be formed?
(c) Part B perform the same function with part M.
  (i) What is M.
  (ii) What is that function it perform?

9. (a) Draw a well labeled diagram of the Mushroom.
(b) In w hat Kingdom does it (mushroom) belong?
(c) List down two uses of the organisms in the kingdom in (b) above.

10. A food chain is given below:
OAK TREE→INSECT LARVAE →INSECTIVOROUS BIRD
(a) A food chain may be represented in a diagram called a "pyramid of number"



Among the pyramids of number A – D; which of these will best represent this food chain?
(b) which is the producer in the food chain?
(c) Name two main groups of organisms termed to as decomposers.

11. (a) What do you understand by the term 'organic evolution'?
    (b) Briefly explain Cosmozoan Theory and Steady-state Theory as explained in theories of origin of life.

12. Describe the adaptation of mammal living in temperate regions.

SECTION C (25MARKS)

Answer **ONLY two (2)** questions from this Section: question**13** is **compulsory.**

13. You are a health officer in Meru District which is infected with Malaria. Educate people on the Malaria by using the following headings in your discussion:
    - Causes of Malaria
    - Transmission
    - Symptoms
    - Effects of Malaria in Man
    - Control of the disease

14. Write an essay on the Economic Importance of Insects.

15. Surfaces where useful materials enter the body of an organism are usually modified to allow maximum surface area for entry of materials.
    Giving one example in plants and two examples in Mammals, explain how an increased surface area may be realized.